MINI-PROJECT 2
LOGBOOK

GROUP MEMBERS

1. Keshav Sharma (124A1150)
2. Khalid (124A1146)
3. Pranjal Wadalkar (124A1181)
4. Milind Sahu (124A1139)

Supervisor

Prof. Reena Gharat



Department of Computer Engineering

SIES Graduate School of Technology

Nerul, Navi Mumbai - 400706

University of Mumbai

(AY 2026-27)

INSTITUTE VISION & MISSION

VISION:

To be a centre of excellence in Education and Technology committed towards Socio-Economic advancement of the country"

MISSION:

1. To impart advanced knowledge in Engineering and Technology.
2. To transform young minds towards professional competence by inculcating values and developing skills.
3. To promote research and ensure continuous value addition among students and employees
4. To strengthen association with industry, research organizations and alumni to enhance knowledge on current technologies.
5. To promote next generation technocracy and nurture entrepreneurial culture for social-economic growth.

DEPARTMENT VISION & MISSION

VISION:

To be a centre of Excellence in Computer Engineering to fulfill the rapidly growing needs of the Society.

MISSION:

1. To Impart quality education to meet the professional challenges in the area of Computer Engineering.
2. To create an environment for research, innovation, professional and social development.
3. To nurture lifelong learning skills for achieving professional growth.
4. To strengthen the alumni and industrial interaction for overall development of students.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Practice Computer engineering in core and muti-disciplinary domains.
PEO2: Exhibit leadership skills for professional growth.
PEO3: Pursue higher Studies for career advancement.

PROGRAMME OUTCOMES (POs)

PO's	OUTCOMES
PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAMME SPECIFIC OUTCOMES (PSOs)

PSO1: To apply computational and logical skills to solve Computer engineering problems.

PSO2: To develop interdisciplinary skills and acquaint with cutting edge technologies in software industries.

STUDENT INFORMATION

Project Title: FootballIQ – AI-Powered Football Analytics & Tactical Intelligence Platform

	Student 1	Student 2	Student 3	Student 4
PRN NO	124A1150	124A1146	124A1181	124A1139
Name	Keshav Sharma	Khalid	Pranjal Wadalkar	Milind Sahu
Class with Division	TE CE (Div E)	TE CE (Div E)	TE CE (Div E)	TE CE (Div D)
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INSTRUCTIONS TO STUDENTS:

1. The logbook must be submitted to the Guide or Co-Guide for verification and evaluation of project activities atleast once in a week.

2. Log book duly signed by guide must be submitted with project report for evaluation at the end of semester to the department

DECLARATION

I declare that this project represents my ideas in my own words and wherever others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my project work. I promise to maintain minimum 75% attendance, as per the University of Mumbai norms. I understand that any violation of the above will be cause for disciplinary action by the Institute.

Yours faithfully

1. **Keshav Sharma**
2. **Khalid**
3. **Pranjal Wadalkar**
4. **Milind Sahu**

(Signature of Students)

Letter of Acceptance

I undersigned, **Prof. Reena Gharat** working in the department of Computer Engineering, willing to guide the project titled **FootballIQ – AI-Powered Football Analytics & Tactical Intelligence Platform** for the mini project 2 Semester V respectively for the academic year 2026-27. The names of the students are:

1. **Keshav Sharma 124A1150**
2. **Khalid 124A1146**
3. **Pranjal Wadalkar 124A1181**
4. **Milind Sahu 124A1139**

_____	_____	_____
Prof. Reena Gharat	**Prof. Sunil K Punjabi**	**Dr. Aparna Bannore**
Mini-Project Guide	**Mini Project Coordinator**	**Head of Department**

COURSE OUTCOMES

CO Number	Course Outcome
CO1	Identify and formulate problems based on societal needs and relevant research.
CO2	Apply engineering knowledge and skills collaboratively to develop solutions for identified problems.
CO3	Demonstrate interpersonal, teamwork, leadership, and self-learning skills during project development.
CO4	Analyze and interpret theoretical, experimental, or simulation results to draw appropriate conclusions.
CO5	Evaluate the societal and environmental impact of solutions while considering sustainability and standard engineering practices.
CO6	Demonstrate effective written and oral communication and apply project management principles during project execution.

Consolidated mapping

CO	POs	PSOs	WPs	EAs	WKs
CO1	PO2, PO3	PSO1, PSO2	WP3, WP6	EA2, EA3, EA4, EA5	WK8
CO2	PO1, PO3, PO5	PSO1, PSO2	WP1, WP2, WP7	EA1, EA2, EA5	WK2, WK4, WK6
CO3	PO8, PO9, PO10	—	WP6	EA1	WK6, WK8, WK9
CO4	PO4, PO2	PSO1, PSO2	WP1, WP3	EA2, EA5	WK2, WK8
CO5	PO6, PO3, PO7	—	WP2	EA3, EA5	WK5, WK7
CO6	PO9, PO10, PO8	—	WP6	EA1	WK6, WK9

CO-PO-PSO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1		3	2	2		2	2				2	2	2
CO2	3	2	3		3			2		2		3	3
CO3								3	2	2	2		
CO4		2		3	2						2	3	3
CO5		2	2			3	2			2			
CO6								2	3	3	2		

ATTRIBUTE DEFINITIONS

Type	ID	Definition
WK	WK1	Systematic, theory-based understanding of applicable natural sciences and awareness of relevant social sciences.
	WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics, and formal computer/information science.
	WK3	Systematic, theory-based formulation of engineering fundamentals.
	WK4	Engineering specialist knowledge and theoretical frameworks for accepted practice areas.
	WK5	Resource efficiency, environmental impacts, whole-life cost, reuse, net-zero carbon, and sustainable operations/design.
	WK6	Knowledge of engineering practice and technology.
	WK7	Role of engineering in society, public safety, and sustainable development.
	WK8	Current research literature, critical thinking, and creative approaches to emerging issues.
	WK9	Ethics, inclusion, professional responsibilities, and norms of engineering practice.
WP	WP1	Requires in-depth engineering knowledge at WK3/WK4/WK5/WK6/WK8 and a fundamentals-based first-principles analytical approach.
	WP2	Involves wide-ranging or conflicting technical, engineering, and other requirements.
	WP3	Has no obvious solution; requires abstract thinking, originality, and analysis to formulate suitable models.
	WP4	Involves infrequently encountered issues.
	WP5	Falls outside problems encompassed by standards and codes of practice.
	WP6	Involves diverse stakeholder groups with widely varying needs.
	WP7	High-level problem includes many component parts or sub-problems.
EA	EA1	Range of Resources: diverse resources including people, money, equipment, materials, and technologies.
	EA2	Level of Interactions: resolving significant interactions between wide-ranging or conflicting issues.
	EA3	Consequences to Society & Environment: significant civic, commercial, economic, ecological, or public-health/safety consequences.
	EA4	Familiarity: unique, novel, or infrequently encountered extension of engineering activities.
	EA5	Judgment: significant independent judgment and creative decisions in unclear or data-scarce situations.

SCHEDULE FOR MINI PROJECT

Date	Week	Contents	Remark	Guide Sign
08/08/2026	1	Project domain selection, team setup, and repository initialization		
14/08/2026	2	Literature survey on computer vision trackers and dataset identification		
21/08/2026	3	Video streaming ingestion pipeline and YOLO detector interface setup		
28/08/2026	4	Pitch boundary definition and HSV spatial field polygon filtering		
04/09/2026	5	Ground-contact coordinate extraction and perspective homography design		
11/09/2026	6	Multi-frame ball detection testing and occlusion handling heuristics		
18/09/2026	7	Ball tracking state machine (detected, possession, unavailable) implementation		
25/09/2026	8	Jersey color classifier using upper torso HSV hue and saturation thresholds		
02/10/2026	9	ByteTrack multi-object tracker integration with low-score detection association		
09/10/2026	10	TeamCalibrator, TeamPersistence temporal filter, and possession engine		
16/10/2026	11	Apache Parquet columnar export, Flask web dashboard, and async job queue		
20/10/2026	12	System integration, end-to-end testing, documentation, and report preparation		

PROGRESS/ATTENDANCE REPORT

Title of the Project: **FootballIQ – AI-Powered Football Analytics & Tactical Intelligence Platform**	
Group No.**3**	Name of Student 1: **Keshav Sharma**
	Name of Student 2: **Khalid**
	Name of Student 3: **Pranjal Wadalkar**
	Name of Student 4: **Milind Sahu**
Supervisor: **Prof. Reena Gharat**	

Sr. No	Date	Attendance				Progress / Suggestion	Mapping		
		1	2	3	4		CO	PO	PSO
1	08/08/2026					Finalized problem statement on automated football match analysis; initialized repository structure.	CO1	PO2	PSO1
2	14/08/2026					Surveyed 12 IEEE/CVF papers on player tracking (ByteTrack, YOLO, SoccerNet); formulated system architecture.	CO1	PO2	PSO2
3	21/08/2026					Built centralized config.py and streaming video reader/writer wrappers using OpenCV; integrated YOLO model.	CO2	PO3	PSO1
4	28/08/2026					Implemented FieldFilter polygon mask to eliminate out-of-pitch spectators and bench detections.	CO2	PO1	PSO1
5	04/09/2026					Designed bottom-center foot coordinates extraction and perspective transformation framework.	CO2	PO5	PSO2

Sr. No	Date	Attendance				Progress/Suggestion	Mapping		
		1	2	3	4		CO	PO	PSO
6	11/09/2026					Formulated ball detection unit tests; addressed intermittent ball detection and bounding box filtering.	CO2	PO3	PSO1
7	18/09/2026					Engineered BallTracker state machine tracking detected, player_possession, and unavailable ball states.	CO2	PO5	PSO1
8	25/09/2026					Developed JerseyClassifier using HSV torso cropping, dominant color masks, and brightness filtering.	CO2	PO3	PSO1
9	02/10/2026					Integrated ByteTrack multi-object tracker; resolved ID swaps across player occlusions and crossings.	CO4	PO4	PSO1
10	09/10/2026					Constructed TeamCalibrator, TeamPersistence temporal filter, and Euclidean distance possession tracker.	CO4	PO5	PSO2
11	16/10/2026					Implemented Apache Parquet columnar data exporter and responsive Flask web interface with async job queue.	CO5	PO3	PSO2
12	20/10/2026					Conducted end-to-end video pipeline verification, performance analysis, project logbook, and final report.	CO6	PO9	PSO1

Sign of the Supervisor

EXAMINER'S FEEDBACK FORM

Name of External examiner:_____

College of External examiner:_____

Name of Internal examiner:_____

Date of Examination: No. of students in project team:

Availability of separate lab for the project: Yes /No

Student Performance Analysis (Put Tick as per your Observation)

Sr. No.	Observation	Excellent (3)	Very Good (2)	Good (1)
1	Quality of problem and Clarity			
2	Innovativeness in solutions			
3	Cost effectiveness and Societal impact			
4	Full functioning of working model as per stated requirements			
5	Effective use of skill sets			
6	Effective use of standard engineering norms			
7	Contribution of an individual's as member or leader			
8	Clarity in written and oral communication			
9	Overall performance			

- Can same mini project extend to next semester by adding new objectives/ideas? (Yes/ No)
- If yes, suggest new Innovative Technique/Idea/ objectives related to this project.

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Signature of External Examiner **Signature of Internal Examiner**